

Indian Institute of Technology Dharwad

CS201L: Artificial Intelligence Laboratory

Lab 13: Hierarchical and DBSCAN clustering

Problem Statement

You are given a dataset file `tsne_2d_features_6_languages.csv` containing 2-dimensional t-SNE features extracted from speech audio embeddings. The dataset consists of 2500 audio samples for each of six different languages (English, Hindi, Kannada, Marathi, Bengali, Manipuri).

Columns 1 and 2 (`tsne_1` and `tsne_2`) represent the 2D feature coordinates. Column 3 (`language`) is the ground truth class label associated with each audio sample. The task here is to partition (cluster) the 2D feature data using Hierarchical (Agglomerative) clustering and DBSCAN clustering, and evaluate the quality of the clusters. While performing the clustering, ignore the third column (`language`) of the data. Use the class information in the third column *only* for computing the evaluation metrics.

1. Data Loading and Exploration

- Load the audio features dataset from the CSV file.
- Separate the features (`tsne_1`, `tsne_2`) and the true labels (`language`).
- Visualize the dataset using a scatter plot shown in figure 1, assigning a distinct color to each of the 6 languages.

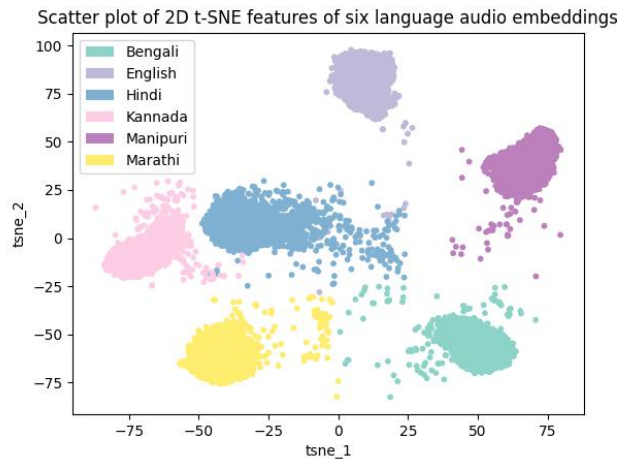


Figure 1: Scatter plot of 2D t-NSE feature dataset with language label

2. Hierarchical Clustering

- Load the dataset `tsne_2d_features_6_languages.csv` and extract the 2D feature dataset.
- Apply Agglomerative Hierarchical Clustering with $n_clusters = 6$ and linkage = "ward" on the dataset. Assign cluster labels to each data point and visualize the clusters using different colors.

- (c) Compute the **purity score**, **Normalized Mutual Information (NMI)**, and **Silhouette score** for this clustering (used linkage = "ward").
- (d) Apply Agglomerative Hierarchical Clustering with $n_clusters = 6$ and linkage = "complete" on the dataset. Assign cluster labels to each data point and visualize the clusters using different colors.
- (e) Compute the **purity score**, **Normalized Mutual Information (NMI)**, and **Silhouette score** for this clustering (used linkage = "complete").
- (f) Apply Agglomerative Hierarchical Clustering with $n_clusters = 6$ and linkage = "average" on the dataset. Assign cluster labels to each data point and visualize the clusters using different colors.
- (g) Compute the **purity score**, **Normalized Mutual Information (NMI)**, and **Silhouette score** for this clustering (used linkage = "average").
- (h) Apply Agglomerative Hierarchical Clustering with $n_clusters = 6$ and linkage = "single" on the dataset. Assign cluster labels to each data point and visualize the clusters using different colors.
- (i) Compute the **purity score**, **Normalized Mutual Information (NMI)**, and **Silhouette score** for this clustering (used linkage = "single").
- (j) Compare all linkage methods based on the computed metrics and identify the best linkage method. Plot the dendrogram using this linkage method to visualize the hierarchical clustering process using `scipy.cluster.hierarchy` functions such as `linkage` and `dendrogram`.

Further, analyze the dendrogram using multiple **threshold levels**. Plot horizontal lines at fixed threshold values of **50** and **100**. In addition, compute the **maximum linkage distance** from the dendrogram and determine another threshold using the thumb rule as **60% of this maximum value**. Plot a horizontal line at this computed threshold as well.

Analyze how many clusters are formed at each of these three cut-off levels (50, 100, and 60% of maximum linkage distance) and compare the results.

- (k) Compare these results with those obtained using **K-means** and **K-Medoids clustering** (performed in the previous lab). Present the comparison in a **tabular format**, include the metrics such as **Purity Score**, **Normalized Mutual Information (NMI)**, and **Silhouette Score** for each method. Based on the table, briefly analyze and identify which clustering approach performs best overall.

3. DBSCAN Clustering

- (a) Load the dataset `tsne_2d_features_6_languages.csv` and extract the 2D feature dataset.
- (b) Consider multiple values for `min_samples` from the set {5, 10, 15, 20, 25}.
- (c) For each chosen `min_samples` value, experiment with `eps` values from the set {2, 3, 5, 6, 7, 8}.
- (d) Apply **DBSCAN clustering** on the dataset for all combinations of `eps` and `min_samples`. Assign cluster labels to each data point.
- (e) For each combination, determine and display:

- Number of clusters formed
 - Number of noise points (labelled as -1)
- (f) Compute the following evaluation metrics (only if more than one cluster is formed):
- **Purity Score**
 - **Normalized Mutual Information (NMI)**
 - **Silhouette Score**
- (g) Visualize the clustered data using a scatter plot:
- Use different colors for different clusters
 - Represent noise points (-1) using **dot markers** (.) instead of circular markers
- (h) Analyze the effect of different `eps` and `min_samples` values on clustering performance and identify the best parameter combination.
- (i) Compare these results with those obtained using **K-means** and **K-Medoids clustering** (performed in the previous lab) and **Agglomerative Hierarchical Clustering**. Present the comparison in a **tabular format**, include the metrics such as **Purity Score**, **Normalized Mutual Information (NMI)**, and **Silhouette Score** for each method. Based on the table, briefly analyze and identify which clustering approach performs best overall.

Note

1. Min sample =Min Point discussed in class
2. Please upload the completed Jupyter Notebook. Copy all Jupyter Notebooks and data files in a folder and rename the folder as your `rollnumber_Lab13`, then create the zip file with the name `rollnumber_Lab13.zip`. Finally, upload the zip file to Moodle for evaluation.

References

- Scikit-learn Agglomerative Clustering documentation: <https://scikit-learn.org/stable/modules/generated/sklearn.cluster.AgglomerativeClustering.html>
- Scipy Hierarchical Clustering (linkage, dendrogram) documentation: <https://docs.scipy.org/doc/scipy/reference/cluster.hierarchy.html>
- Scikit-learn DBSCAN documentation: <https://scikit-learn.org/stable/modules/generated/sklearn.cluster.DBSCAN.html>
- Scikit-learn Nearest Neighbors (for distance computation): <https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.NearestNeighbors.html>

Functions / Code Snippets

```
# Code snippet for k mean clustering.
import pandas as pd

from sklearn.cluster import AgglomerativeClustering, DBSCAN
from sklearn.metrics import normalized_mutual_info_score, silhouette_score,
↳ adjusted_mutual_info_score

from sklearn.metrics import normalized_mutual_info_score, silhouette_score #
↳ Compute NMI and Silhouette Score
import matplotlib.pyplot as plt # Visualization of clusters

# 1. Load data
tsne_df = pd.read_csv("tsne_2d_features_6_languages.csv")

# 2. Apply Hierarchical Clustering

agg_model = AgglomerativeClustering(n_clusters=6, linkage='ward')
agg_labels = agg_model.fit_predict(X)

# 3. Compute evaluation metrics for Hierarchical Clustering
nmi = normalized_mutual_info_score(y_true, agg_labels)
sil_score = silhouette_score(X, agg_labels)

# 4. Plot dendrogram
sch.dendrogram(linked, truncate_mode='level')

# 5. Threshold at 55
plt.axhline(y=55, color='green', linestyle='--', label='Threshold = 55')

# 6. Threshold at 80% of max distance
max_dist = np.max(linked[:, 2])
threshold_80 = 0.8 * max_dist

# 7. Apply DBSCAN Clustering

dbscan_model = DBSCAN(eps=3.0, min_samples=4)
dbscan_labels = dbscan_model.fit_predict(X)

# 8. Compute evaluation metrics for DBSCAN Clustering
nmi = normalized_mutual_info_score(y_true, dbscan_labels)
sil_score = silhouette_score(X, dbscan_labels)
```